Minimizing Computational Entropy via Physical Power-Gate Control

A Proposal for Dynamic Power Control and Hallucination Suppression in Existing Computing Infrastructure

- Logical Execution of Arithmetic Gates through Algebraic Exclusion Laws -

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Abstract

By 2026, the power consumption driven by the exponential proliferation of Generative AI is projected to surpass 1,000 TWh. Consequently, local power infrastructures housing data centers are confronting physical thermodynamic limits. Existing probabilistic inference (152-Legacy) continuously dissipates massive "logical friction heat" to achieve statistical convergence; conventional symptomatic optimizations struggle to bypass this increasing entropy.

This paper proposes a solution based on the "AG-UPL" framework—redefining computational structures by identifying cumulative errors in existing mathematics through the 163-Standard (Original OS) metrics. We shift the computational basis from "probability" to "consistency." The core of this method lies in applying Algebraic Exclusion Laws via ν_2-valuation to detect logical inconsistencies prior to execution. This enables "Logical Gating"—the dynamic suspension of power to inconsistent arithmetic paths by leveraging existing clock-gating functions in current processors.

Theoretical verification demonstrates that this protocol simultaneously achieves significant effective power reduction and suppression of hallucinations (logical contradictions) by dynamically restricting power to inconsistent paths within existing GPU/NPU infrastructures. This represents a pragmatic, immediately deployable engineering proposal to transform AI from a resource consumer into an energy-preserving entity through logical refinement.

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Chapter 1: Introduction — Computational Entropy and the Thermodynamic Limits of Power Infrastructure

1.1 Physical Constraints of Power Supply and Demand in 2026

As of 2026, the surge in computational demand for Generative AI has reached a point of conflict with the supply capacity and thermal design limits of existing power infrastructure.

Power Consumption Projections Based on IEA Statistics: According to the 2024 IEA (International Energy Agency) report, global electricity consumption by data centers (DCs), cryptocurrency, and AI is expected to reach 1,000 TWh by 2026. This scale is comparable to the total annual power generation of Japan (approx. 900–1,000 TWh).

Risk of Local Power Grid Collapse: In certain DC hubs in Ireland and North America, DC-driven power demand has begun to exceed 20–30% of total demand. A physical "grid bottleneck" is emerging where the expansion of transmission networks cannot keep pace with the growth in demand.

Intelligence Energy Deficit: While a traditional keyword search consumes approximately 0.3 Wh per query, a single LLM inference is estimated to consume between 3.0 Wh and 10.0 Wh. This paper defines the current state—where energy input is 10 to 30 times higher without a commensurate increase in the "qualitative conversion efficiency" of information—as an "energy deficit in intelligence."

1.2 Physical Definition of Computational Entropy and Waste Heat

The primary driver of power consumption in the existing "152-Legacy" models is the increase in "computational entropy" associated with irreversible information processing.

Landauer's Principle (Physical Limits): According to Landauer's Principle, erasing or irreversibly changing one bit of information generates a minimum of $kT \ln 2$ heat (where k is the Boltzmann constant and T is the absolute temperature).

Occurrence of Logical Friction: Probabilistic inference processes—such as calculating probability distributions via softmax functions and computing massive unnecessary paths—ultimately discard vast amounts of intermediate results to select a single solution. This irreversible destruction of information is converted into physical heat (computational entropy) within CPU/GPU circuits.

Hallucinations as Negative Entropy: The generation of logically inconsistent data (hallucinations) produces physical heat without generating useful information. In terms of system physics, this is synonymous with the emission of "negative entropy," which nullifies the input energy.

1.3 Engineering Limits of Existing Optimization Methods

Commonly adopted "model quantization" or the introduction of specialized hardware remain symptomatic treatments that preserve the underlying logical inconsistencies.

Unaddressed Cumulative Errors: Leaving the cumulative errors inherent in existing floating-point arithmetic (defined in this paper as an 11-bit deficit) unaddressed leads to an exponential increase in both hallucinations and power consumption as models scale.

Direction of this Proposal: This paper presents a method to logically discard inconsistent arithmetic paths before they reach the physical layer (the power-conveying layer).

1.4 "Attempted" Computation via Logical Gating

While existing inference processes rely on "brute-force matrix operations," this method verifies data attributes (parity) before execution. Arithmetic paths confirmed to have attribute mismatches are halted at the hardware gate level (Logical Gating). This intercepts physical energy consumption before computational entropy is generated.

Chapter 2: Fundamental Principles — Arithmetic Control via Algebraic Exclusion Laws

2.1 Redefining Computational Bases: Returning to the Original OS (163-Standard)

The "11-bit deficit" in the 152-Legacy system is defined as the difference (Δ11) between the largest Heegner number d=163 (where the class number of the imaginary quadratic field $\mathrm{Q}(\sqrt{-d})$ is 1) and the existing standard arithmetic basis (152).

Information Saturation and Voids: Compared to the 163-bit logical space, which maintains the densest mathematical consistency, the existing 152-bit equivalent structure lacks the capacity to hold 11 bits of logical information. This "void" accumulates as rounding errors in floating-point operations and micro-entropy in high-dimensional matrix products.

Degradation of Encoding Efficiency: To compensate for this 11-bit deficit, existing LLMs require statistical redundancy (massive parameter scaling). Because they lack the deterministic consistency of the 163-Standard, the 152-Legacy systems must iterate trillions of probabilistic approximation calculations, which is the direct physical cause of thermal loss.

2.3 Pre-execution Detection of Inconsistency via ν_2-Valuation (2-adic Valuation)

The core technology for anchoring the logic of the Original OS (163-Standard) to the physical layer is the pre-execution "censorship" of instructions using ν_2-valuation.

Application of Algebraic Exclusion Laws: Every arithmetic instruction is inspected for its parity and algebraic consistency before execution.

Deterministic Bypass: If an arithmetic result is predicted to be inconsistent (even-numbered noise) relative to the 163-Standard, the corresponding arithmetic path is physically prevented from executing. This shifts the process from "verifying errors after computation" to "discarding inconsistencies before execution."

2.4 Power-Gate Control at the Physical Layer: Logical Gating

This method does not assume the development of new hardware; instead, it executes physical power saving by logically controlling the low-level functions of existing processors.

Instruction Substitution (NOOP Substitution): Instructions judged as non-compliant during the consistency check (v_2-valuation) are dynamically replaced with "No Operation" (NOOP) instructions or instructions that maintain a low-power state just before reaching the Arithmetic Logic Unit (ALU).

Inducing Clock Gating: For arithmetic blocks judged logically invalid, the clock supply to the processor is temporarily intercepted. This suppresses the physical activity factor through logical control from the software side.

2.5 Dynamic Suppression of Hallucinations

Hallucinations are "physical error signals" output as a result of logical inconsistencies being permitted to draw power and pass through arithmetic circuits. Under Logical Gating via AG-UPL, inconsistent logical paths are denied power; thus, hallucinations are eliminated at the physical layer before they can be generated.

Chapter 3: Theoretical Verification — Mechanism of Arithmetic Step Reduction and Power Control

3.1 Shortest-Path Optimization via Elimination of Statistical Trials

This method minimizes the "transfer resistance" of intelligence by eliminating unnecessary statistical trials in the inference process. Existing AI requires massive matrix operations because the logical paths to a solution are probabilistically dispersed. AG-UPL uses parity consistency (163-Standard) as a filtering metric to deterministically optimize and shorten the arithmetic route.

3.2 Pre-execution Evaluation via Consistency Function Φ

Before each Arithmetic Logic Unit (ALU) processes data x and weight W, the following consistency function Φ is applied to evaluate the validity of continuing the operation.

$$\Phi(x, W) = v_2(x \oplus W)(\ mod \text{ Parity_Constant})$$

$\boldsymbol{v_2(n)}$**:** The 2-adic valuation; the exponent of the highest power of 2 that divides the integer n.

Parity_Constant: A constant defined based on the Original OS (163-Standard) to ensure logical integrity.

If the return value of Φ indicates "inconsistency" (even-numbered noise), the operation is judged invalid. Since this determination is completed via lightweight bitwise operations (XOR and valuation check), it is possible to discard the instruction itself before executing heavy floating-point operations.

3.3 Logical Suppression Model for Dynamic Power Consumption (P_{dyn})

Dynamic power consumption P_{dyn} in existing CMOS processors is defined by the following equation:

$$P_{dyn} = \sum_{i=1}^{n} (C_i \cdot V^2 \cdot f \cdot \alpha_i)$$

$\boldsymbol{C_i}$**:** Switching capacitance

$\boldsymbol{V}$**:** Supply voltage

$\boldsymbol{f}$**:** Operating frequency

$\boldsymbol{\alpha_i}$**:** Activity factor (the rate at which circuits switch)

AG-UPL directly controls the activity factor α based on the results of the consistency function Φ. For arithmetic threads judged non-compliant, α is approximated to zero by forcing clock gating or substituting instructions with NOOPs.

3.4 Arithmetic Consolidation via "1729: Ramanujan Anchor"

In this method, "1729" functions as a "Logical Anchor" to consolidate multiple independent arithmetic paths into a single consistency check.

Utilization of Mathematical Properties: 1729 is the "taxicab number"—the smallest positive integer expressible as the sum of two cubes in two different ways ($1^3 + 12^3$ and $9^3 + 10^3$). This multi-representational property, where different paths lead to the same solution, is applied as the ultimate form of Common Subexpression Elimination (CSE) in computer architecture.

Arithmetic "Merging": Typically, AI executes similar calculations in separate threads, consuming power individually. By performing parity verification using 1729 as an anchor, structurally common arithmetic groups are handled by reusing a single calculation result.

Physical Benefits: "Packing" 1,000 arithmetic instructions into a single anchor verification drastically reduces the frequency of memory access and register switching. This serves as the third "hemostatic point" (point of interception) to further drive down the activity factor α.

3.5 Case Study: Suppression of Anatomical Hallucinations in Image Generation

The frequent "miscounting of fingers" in image generation (e.g., diffusion models) is caused not by a lack of training data, but by the permission of "inconsistent tensors" (multidimensional arrays) due to cumulative errors during the calculation process.

Identification of Inconsistent Tensors: Arithmetic paths attempting to form anatomically unnatural shapes (e.g., a sixth finger) in high-dimensional vector operations deviate from the parity of the 163-Standard.

Physical Rejection (Tensor Gating): The consistency function Φ detects these "inconsistent vectors" during each sampling step. If the determination is "non-compliant," power to the arithmetic core responsible for that pixel region is physically intercepted.

Engineering Outcome: While the AI retains its correct statistical knowledge (the "5-finger" blueprint), the physical "inconsistent construction" (the 6th finger) is rejected at the gate level. Consequently, the output is forced to converge toward the mathematically stable "5-finger" structure, simultaneously reducing rendering errors and wasted power.

Chapter 4: Social Impact — Resource Redistribution and Decentralization of Intelligence

4.1 Grid Recovery: Direct Contributions to Power Infrastructure

The power saved through the logical optimization of arithmetic processes (estimated at 5% to 15% or more) is immediately returned to the primary power grid as physical surplus electricity.

AI as an Energy Balancer: This method transforms AI from a "resource consumer" into a "regulatory valve" that stabilizes power supply and demand through logical refinement. This is defined as "Grid Recovery," contributing to the alleviation of local power shortages.

4.2 Decentralization of Intelligence and Democratization of Computing via Edge Rendering

Minimizing computational entropy allows AI to break free from its dependence on massive data centers (the cloud), physically enhancing the inference performance of edge devices such as smartphones and PCs.

Performance Liberation under Thermal Design Power (TDP) Constraints: Mobile devices have physical limits on heat dissipation (TDP limits). While high-load matrix operations in the 152-Legacy system trigger thermal throttling (performance degradation due to heat) within seconds, AG-UPL suppresses the heat generation itself by intercepting inconsistent arithmetic paths. This allows devices to maintain their peak performance for extended periods.

Elimination of Memory Bandwidth Bottlenecks: In edge devices, the transfer of massive model parameters is a primary driver of power consumption and latency. Consolidating operations using the 1729 Anchor logically reduces the frequency of memory bus access, enabling high-speed inference even with limited hardware resources.

Securing Offline Autonomy: By improving arithmetic efficiency, advanced inference previously relegated to the cloud can now be completed locally (on-device). This ensures the "continuity of thought" during network disruptions while reducing communication costs and protecting privacy.

4.3 Securing Intellectual Sovereignty and Reducing Verification Costs

Deterministic output based on the Original OS (163-Standard) eliminates the "fluctuation of responses" inherent in probabilistic inference.

Automation of Verification: Since parity consistency is physically guaranteed, the temporal and cognitive costs for humans to "check for hallucinations" are minimized. This is a resource-efficient, autonomous process that allows humans to maintain sovereignty over information.

Chapter 5: Conclusion — Logical Soft Landing in Existing Infrastructure

5.1 Paradigm Shift to "Quality of Arithmetic" (Consistency)

This paper proposes a shift in the value metric of AI from "Quantity" (Scaling Laws) to "Quality" (Parity Consistency). In response to the power supply crisis of 2026, we have presented a process that physically rejects the "noise that consumes power and generates errors," extracting only mathematical truths based on the Original OS (163-Standard).

5.2 Creating a "Strategic Grace Period" via Existing Assets

The most significant feature of this proposal is the ability to optimize current computing assets (GPUs/NPUs, etc.) through the power of logic alone, without waiting for the completion of next-generation semiconductors. Dynamic power control at the instruction-set level suppresses the energy consumption curve for data centers and society as a whole, functioning as a definitive "Strategic Grace Period" to prevent the collapse of the power grid.

5.3 Objective Execution as an Independent Researcher

The "Power-Gate Control via Algebraic Exclusion Laws" presented in this paper is a pure engineering optimal solution that could not arise from existing business models that profit from the wasteful consumption of computational resources. This approach, which returns hardware to its role as a "device for processing truth," is a technical hemostatic measure directly linked to the sustainability of civilization, providable only from the perspective of an independent researcher.

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Finally, I sincerely hope that this study will assist in organizing the computational complexity inherent in "probabilistic inference" and contribute to rebuilding a new arithmetic paradigm based on discrete "Parity Consistency." I hope this research serves as a new starting point for understanding the relationship between information processing and energy consumption from a more substantive perspective.